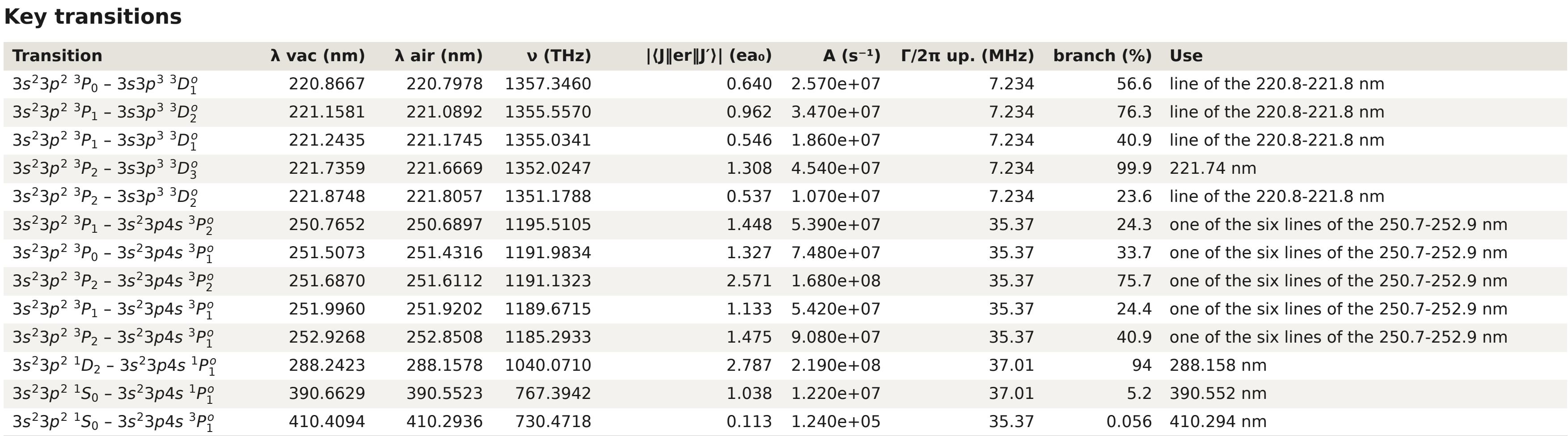


Si (Si I). Strongest classified transitions below 2 μm (NIST): 44 levels, 79 lines.

Si⁺ ionisation limit 65747.76 cm⁻¹ = 8.15168 eV (152.096 nm)

Arrow colour = vacuum wavelength (nm)



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3s^2 3p^2 \ ^3P_1$	1.840	-	F=1/2: -1.84 F=3/2: +0.92
$3s^2 3p^2 \ ^3P_2$	-160.1(13)	-	F=3/2: +240.15 F=5/2: -160.10
$3s3p \ ^3D^o_3$	-532.9(6)	-	F=5/2: +1065.80 F=7/2: -799.35

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3s^2 3p^2 \ ^3P_2 - 3s 3p^3 \ ^3D_3^o$	221.736	29-28	1753.3(11)
$3s^2 3p^2 \ ^3P_2 - 3s 3p^3 \ ^3D_3^o$	221.736	30-28	3359.9(6)
$3s^2 3p^2 \ ^3P_2 - 3s^2 3p4s \ ^3P_2^o$	251.687	30-28	330(30)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$